



Ramrachai Marma

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Work Permit: Available

SUMMARY

Data Scientist with an M.Sc. in Data Science, experienced in developing end-to-end AI and machine learning solutions from data preparation and model development to cloud-based deployment. Skilled in transforming unstructured data into structured, decision-ready information using machine learning, NLP, LLMs, and data engineering approaches. Experienced with Azure-based AI solutions, data pipelines, and human-in-the-loop systems, with a strong ability to connect technical solutions to practical business and engineering needs.

PROFESSIONAL EXPERIENCE

Working Student, Data Science

June 2024 – Present

Siemens Mobility GmbH | Erlangen, Germany

- Analyzed operational and incident data using Qlik Sense and Power BI to identify trends, monitor KPIs, and support engineering decisions.
- Used Python and SQL for data preparation, cleaning, transformation, validation, and analysis of structured and unstructured data.
- Applied machine learning, NLP, and LLM techniques to engineering data for information extraction, classification, and model evaluation.
- Automated data and operational workflows using Python, Power Automate, Azure DevOps, and CI/CD, reducing manual processing.
- Conducted data quality, risk, and gap analyses across cloud and industrial workflows and communicated findings to technical and business stakeholders.

Master's Thesis: AI-Assisted Locomotive Fault Reporting System

January 2025 – July 2026

Siemens Mobility GmbH | Erlangen, Germany

- Architected an end-to-end LLM application converting unstructured German locomotive maintenance logs into validated records using Azure OpenAI, Azure Functions, and Streamlit.
- Designed a dual AI extraction and categorization methodology applying few-shot prompting and mapping free-text reports to a controlled 125-category bilingual engineering taxonomy.
- Engineered an automated layout-aware ingestion pipeline (DATA_Extractor) parsing PDF, DOCX, and JSON engineering documents into a structured JSONL knowledge base via Azure Document Intelligence.
- Implemented a Human-in-the-Loop (HITL) validation framework allowing domain experts to review, edit, and evaluate model outputs prior to database storage, ensuring high extraction accuracy.
- Built a serverless orchestration backend using Azure Functions and Blob Storage to handle LLM calls, historical context retrieval, and automated SharePoint synchronization.

Data Analyst

May 2021 – May 2022

aiQuest Intelligence | Remote

- Applied quantitative data analysis and data validation methodologies to monitor operational KPIs and learner performance on an online platform.
- Developed data cleaning and transformation workflows to deliver structured analytical reports for management decision-making.

Finance Data Analyst Intern

August 2017 – December 2017

Oasis Group | India

- Performed data auditing and discrepancy analysis on financial datasets to ensure reporting consistency and data integrity.

EDUCATION

M.Sc. Data Science

April 2023 – July 2026

Friedrich-Alexander-Universität Erlangen-Nürnberg, Germany

B.Eng. Information Technology

June 2016 – September 2020

Gujarat Technological University, India

SELECTED PROJECTS

- **Enterprise RAG & Knowledge Retrieval Pipeline:** Built a production-ready RAG system utilizing semantic vector search, chunking strategies, and secure access governance for high-precision technical document retrieval.
- **Requirements Specification Extraction Tool:** Developed an automated parsing tool using Azure Document Intelligence and LLMs to extract, normalize, and deduplicate engineering specifications from multi-format files.
- **Automated Data Engineering & LLMops Pipeline:** Implemented an end-to-end Python data pipeline with CI/CD integration for automated, reproducible data transformations and schema validation.
- **Medical Chatbot (Llama 2 RAG):** Built a retrieval-augmented medical Q&A chatbot running a locally-hosted Llama 2 model against a Pinecone vector store using LangChain, served through a Flask web application.
- **Personal Face Recognition System:** Built a system to find a specific person's face across group photos, combining Haar Cascade detection with an LBPH recognizer trained on reference images in Python/OpenCV, producing confidence-scored, annotated output.
- **Cosmetics Brand Analysis:** Conducted a market-research case study on cosmetics consumer data, performing data cleaning, exploratory analysis, Net Promoter Score scoring, and NLP sentiment analysis (Pandas, Seaborn, Scikit-learn, NLTK/VADER) on open-text feedback to produce segmentation and marketing recommendations.
- **Automated Weather Forecast Generator:** Developed a Python automation pulling live data from the OpenWeatherMap API for 15 cities across the UK, India, and the US, and rendering it into styled, date-stamped PNG/PDF forecast images using Pillow.

SKILLS

Methodological & AI Skills: RAG, agentic workflows & tool-calling, Model Context Protocol (MCP) & MCP server development, few-shot prompt engineering & evaluation, human-in-the-loop (HITL) validation, layout-aware document parsing, semantic search & taxonomy mapping, quantitative data analysis, risk assessment & gap analysis, Agile/Scrum

Machine Learning & GenAI: Large language models (LLMs), Azure OpenAI, Azure Document Intelligence, Azure AI Foundry, NLP, embeddings & vector search, structured outputs (Pydantic / JSONL)

Programming & Automation: Python, SQL, Power Automate, GitHub Actions CI/CD, Azure DevOps CI/CD, Streamlit, REST APIs, n8n

Cloud & Infrastructure: Microsoft Azure (Azure Functions, Blob Storage), AWS (basic), Docker, Kubernetes.

Data Engineering & Analytics: ETL pipelines, data transformation, MySQL, Snowflake, Power BI, Qlik Sense

Security & Governance: IT security engineering, cloud security, access governance, risk assessment, IT audit

LANGUAGES

English: C1 (Full professional proficiency)

German: B1+ (Working knowledge; incl. German-language NLP/technical datasets)

CERTIFICATES

- Get Started with Python, Google/Coursera (Credential ID: UYKMUAN9VFW2)
- Foundations of Data Science, Google/Coursera (Credential ID: F4ZWYPFMX3CF)
- AWS Cloud Technical Essentials, Amazon Web Services (Credential ID: FIVGF7237XUY)
- Introduction to Linux (Credential ID: UBXEQEKH7EQJ)
- SQL, MySQL for Data Analytics and Business Intelligence, Udemy (Credential ID: UC-f644c030-4c86-4c68-b651-bc1e7745bc95)
- Building Your First AI Agent with OpenAI (Credential ID: 8GD4RJ18J1S4)
- AI for Data Analysis, Google (Credential ID: 25TGXAIWYYBE)

AWARDS

- Fully Funded Government Scholarship (ICCR, Government of India): competitive, merit-based award for study abroad.